EXPERIMENTAL RESULTS & REPRODUCIBILITY REPORT

This document preserves the current Ubuntu-generated figures and records the interpretation of the experimental stage.

1. Experimental Configuration

Five workloads were evaluated using conventional FCFS and carbon-aware FCFS. The carbon-aware scheduler used a threshold of 500 gCO_2 /kWh and a synthetic temporal carbon trace. CPU power was modeled as 50 W.

2. Carbon Emissions

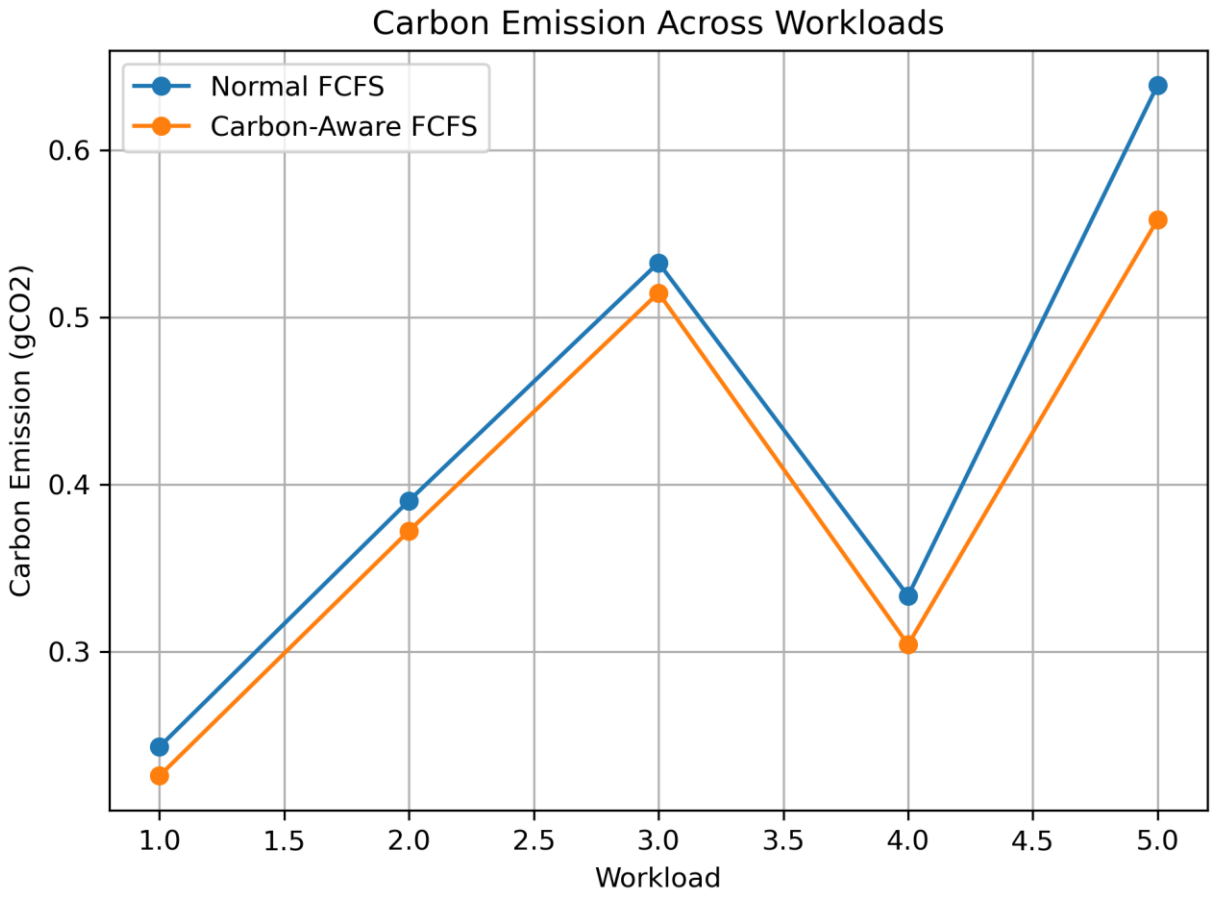


Figure 1. Carbon emissions across five workloads. Carbon-aware FCFS remains below conventional FCFS for every plotted workload.

3. CPU Utilization

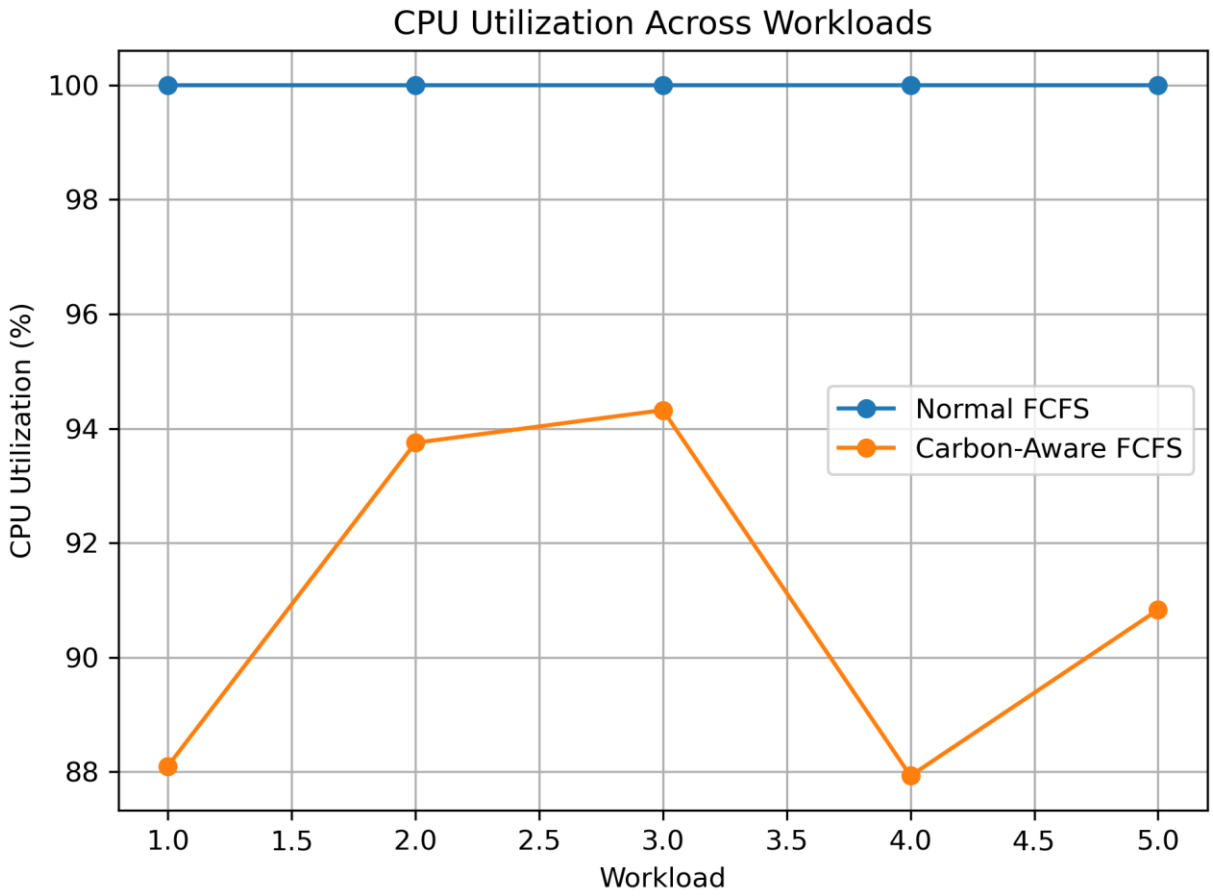


Figure 2. CPU utilization. Lower utilization in carbon-aware FCFS reflects deliberate idle intervals introduced to avoid higher-carbon execution periods.

4. Energy

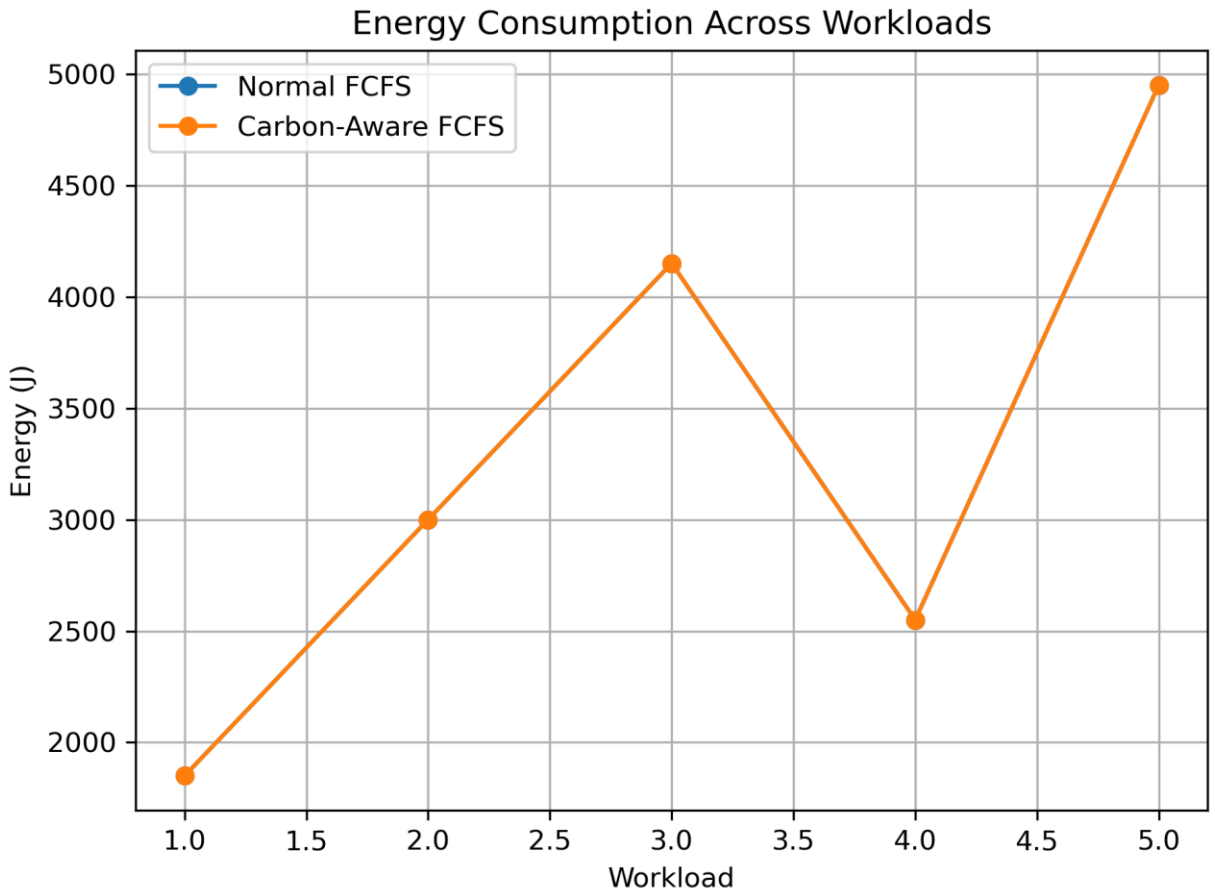


Figure 3. Energy consumption. The plotted series overlap under the constant-power model, demonstrating that the current carbon reduction mechanism operates through temporal relocation rather than reduced CPU energy.

5. Waiting and Turnaround Time

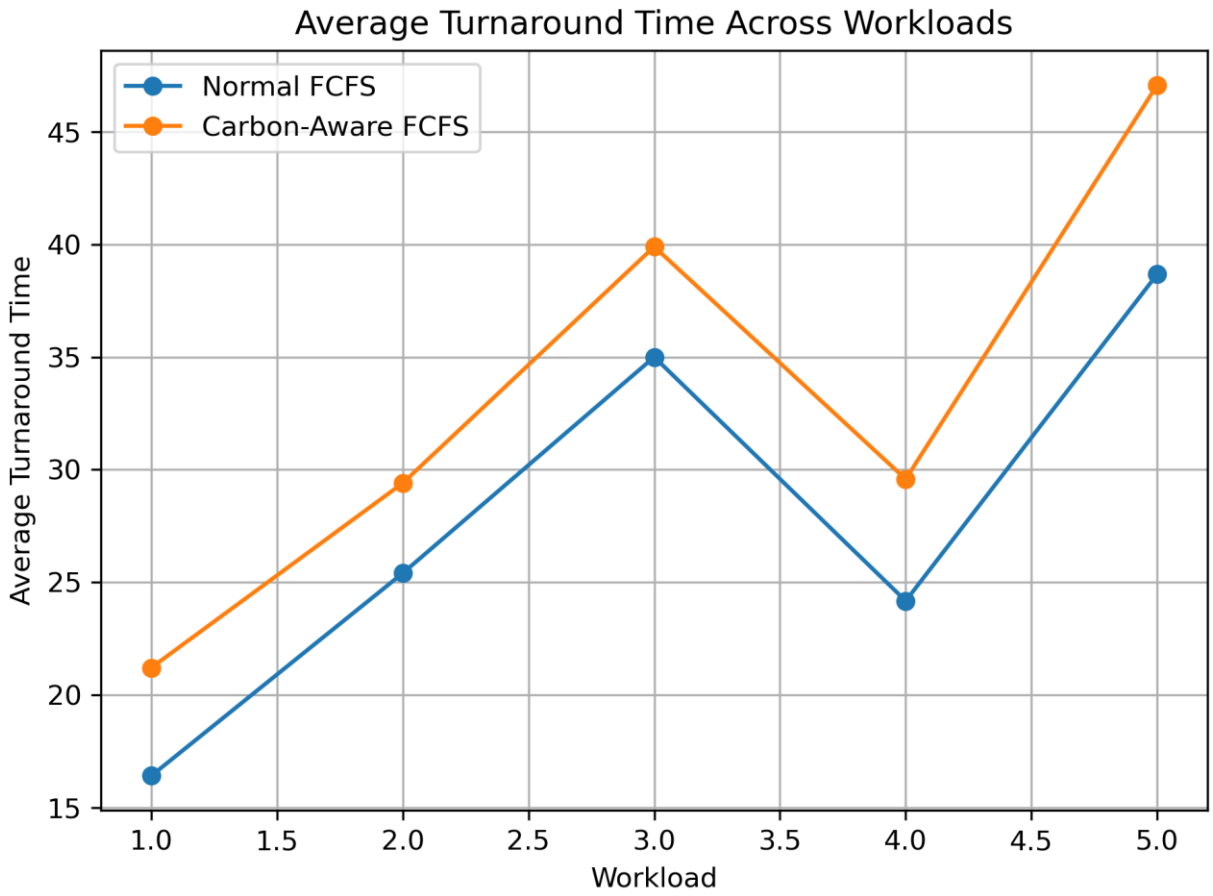


Figure 4. Average turnaround time across workloads.

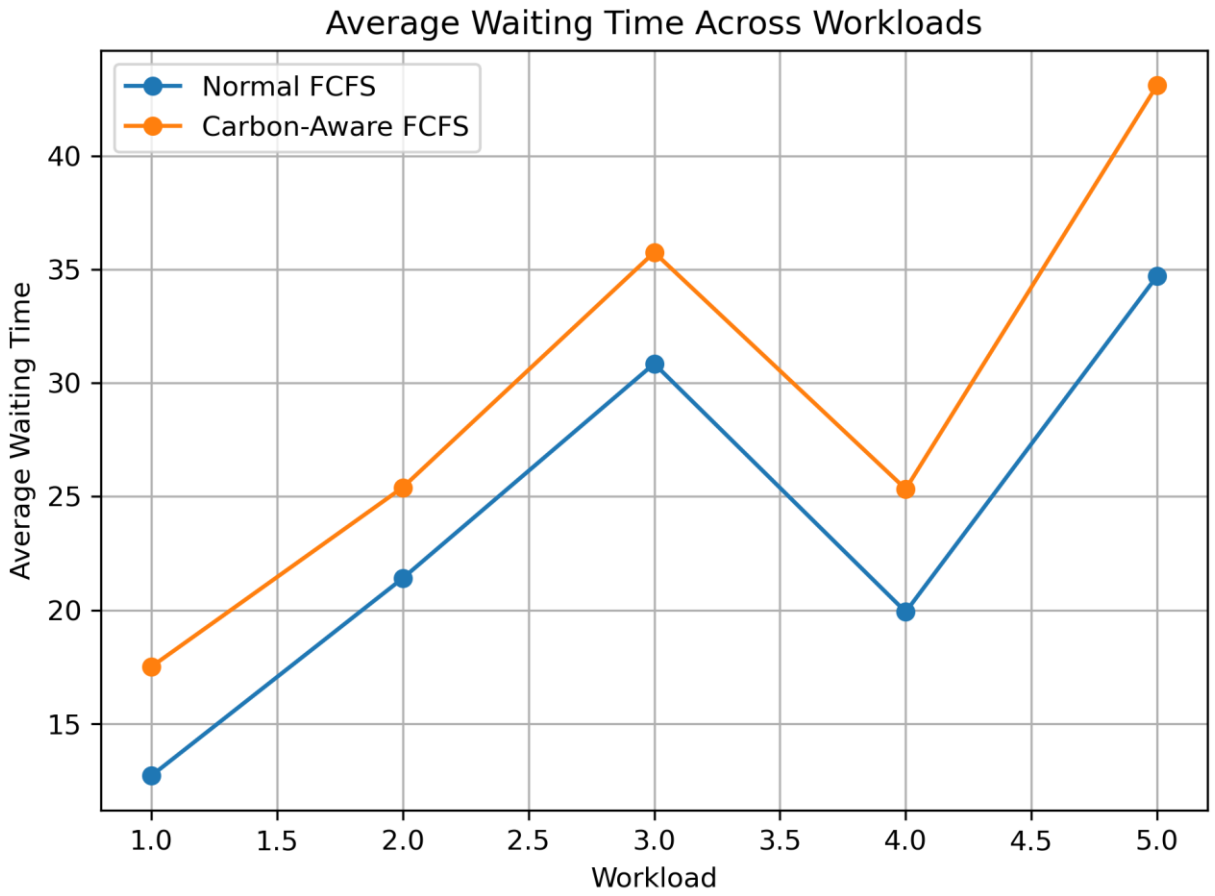


Figure 5. Average waiting time across workloads.

Both latency metrics increase for carbon-aware FCFS because temporal carbon shifting deliberately postpones some processes. This is a measurable cost of carbon reduction and should be evaluated jointly with emissions rather than hidden.

6. Supplementary Aggregate Figures

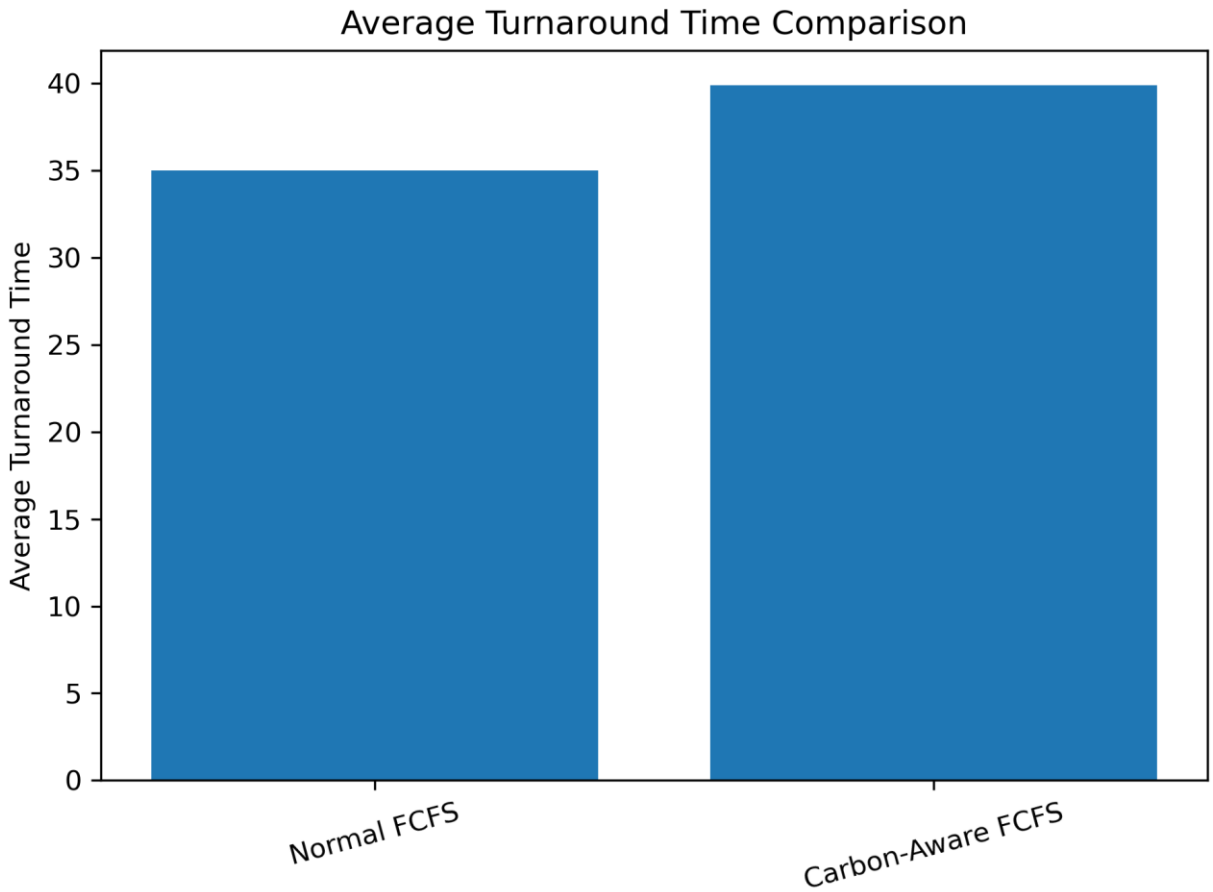


Figure 6. Aggregate average turnaround time comparison.

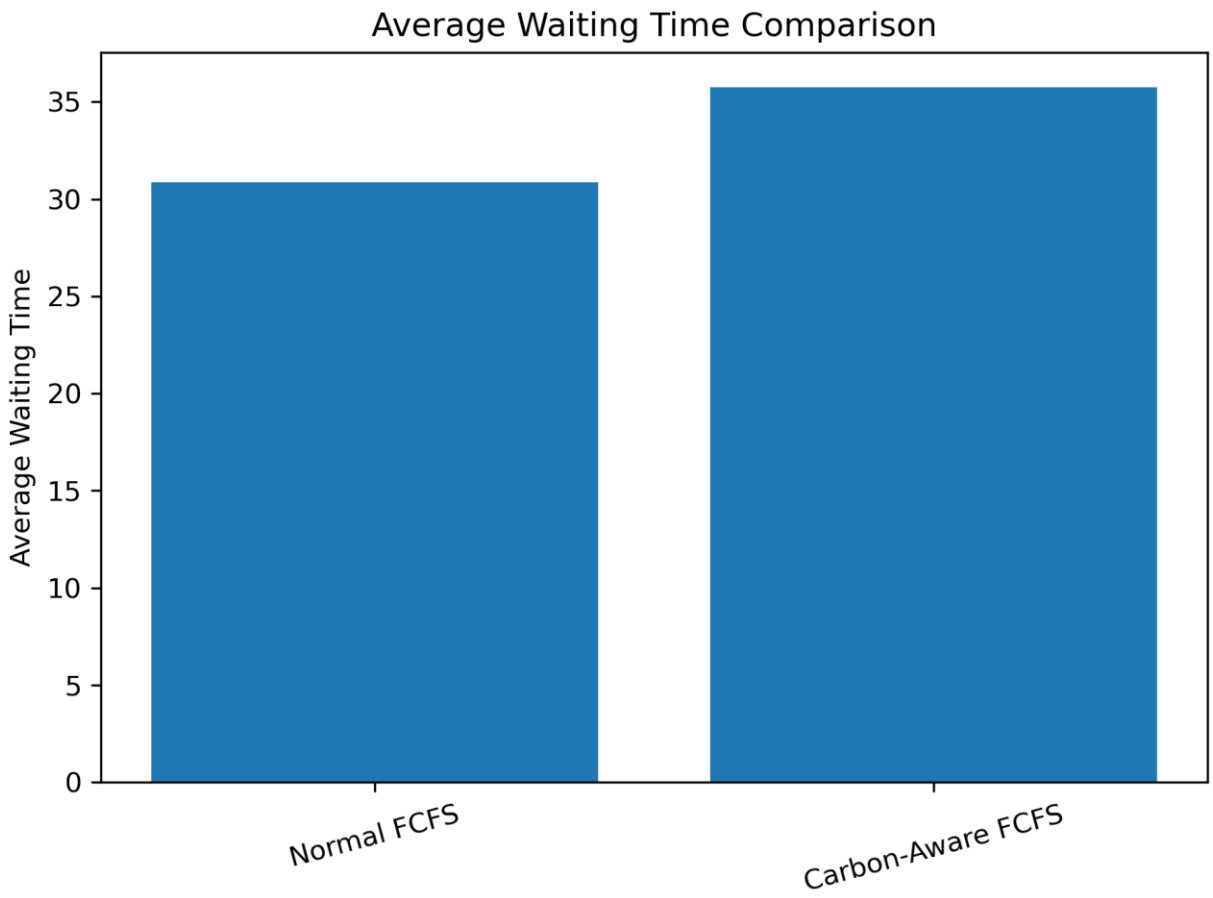


Figure 7. Aggregate average waiting time comparison.

7. Reproducibility Checklist

- Compile with GCC in the Linux project environment.
- Run both baseline and carbon-aware schedulers against the same workload file.
- Retain the carbon trace unchanged for paired comparisons.
- Store raw outputs before parsing or graph generation.
- Generate figures only from the machine-readable results.
- Record compiler version, VM configuration and carbon-data source when the VM stage begins.

8. Interpretation Boundary

Important

These figures establish simulation behavior, not real-world carbon savings. The present results should be described as estimated emissions under the stated synthetic trace and constant-power model. Real-system claims require VM or hardware validation.